

Accelerating phase prediction via CALPHAD-informed machine learning and data augmentation

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ARTICLE INFO

Keywords:

High-entropy alloys
Machine learning
CALPHAD
Phase prediction
Data augmentation

ABSTRACT

The immense compositional space of alloys presents a significant challenge for rapid phase prediction and renders exhaustive experimental exploration infeasible. This work introduces a machine learning (ML) framework that leverages CALPHAD-based thermodynamic modeling to accelerate phase stability predictions. The methodology is demonstrated through FCC phase prediction in a High Entropy Alloy system as a representative case study, with transferability further illustrated through BCC phase prediction using the same database and pipeline. A dataset of ~72,000 compositions within a ten-element compositional space was generated using Thermo-Calc software. To mitigate data scarcity and class imbalance, a novel multidimensional interpolation strategy, grounded in the lever rule, was employed for data augmentation near phase boundaries, substantially expanded the training set without requiring additional CALPHAD calculations. Four ML models were evaluated: KNN, RF, GB and SVM, with the SVM consistently exhibiting superior performance across all configurations. Data augmentation improved balanced accuracy by 1–3%, with gains of up to 6.5 percentage points for the minority single-phase FCC class. The proposed methodology provides a robust, scalable, and computationally efficient strategy for the data-driven design of alloys.

1. Introduction

The exploration of complex compositional spaces demands robust computational approaches, with computational thermodynamics playing an essential role. The CALPHAD (CALculation of PHase Diagrams) method provides a rigorous framework for predicting phase stability in multicomponent systems through extrapolation from critically assessed binary and ternary data [1–6]. This approach enables efficient virtual screening across wide composition and temperature ranges [1,5]. Yet, even with its efficiency, high-throughput CALPHAD calculations remain time-consuming, often requiring days or weeks for comprehensive mapping [2,6,7]. This is especially true for high-entropy alloys (HEAs) [8–10], important candidates for structural applications [9–11], whose vast compositional space [12] created by using multiple principal elements allows extensive property tuning [9].

To address this limitation, machine learning (ML) has recently emerged as a powerful complementary tool, capable of capturing complex, non-linear relationships between composition, processing, and microstructure [13–15]. Integrating ML with thermodynamic modeling offers a promising route to accelerate phase prediction. For instance,

regression surrogate models have been developed to accelerate CALPHAD phase fraction predictions [16–18]. Besides, in alloy design, phase formation is often treated as a classification problem. Classification surrogate models were developed from CALPHAD calculations by Zeng et al. to design single-phase HEAs [19,20]. However, because the class of interest, e.g., single-phase FCC alloys, is often a minority within the compositional space, it may be poorly predicted or even overlooked by the surrogate model. To address this class-imbalance problem, Zeng et al. performed additional CALPHAD calculations, either by sampling a fine grid while retaining only minority-class data [19], or by preferentially sampling regions likely to contain minority classes [20]. Both approaches require additional computation time.

In this work, we propose a general machine learning framework for accelerated phase prediction that integrates CALPHAD thermodynamics with a novel data augmentation strategy. The overall workflow is summarized in Fig. 1, which illustrates the complete data pipeline from CALPHAD thermodynamic calculations through dataset filtering, train/test splitting, data augmentation, machine learning training, and model evaluation. The class imbalance is addressed through multidimensional interpolation guided by the lever rule, generating synthetic

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<https://doi.org/10.1016/j.commsci.2026.114834>

Received 12 January 2026; Received in revised form 28 May 2026; Accepted 29 May 2026

Available online 4 June 2026

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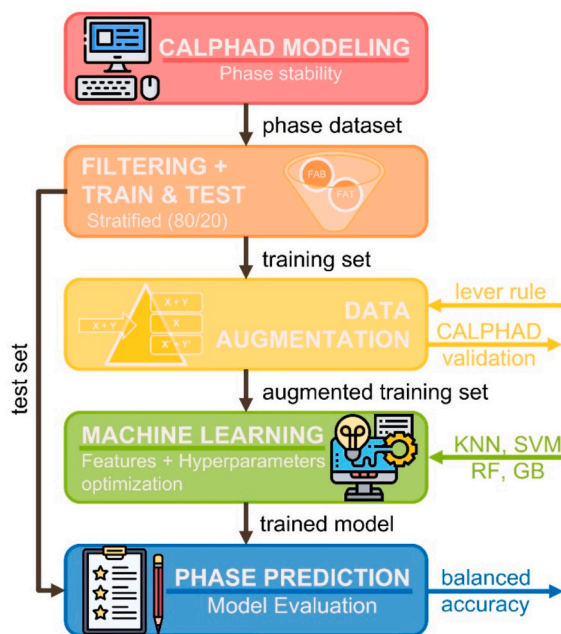


Fig. 1. - Schematic overview of the proposed framework, illustrating the data flow from CALPHAD thermodynamic modeling through FAB/FAT filtering, stratified train/test split, lever-rule-based data augmentation, and machine learning training and evaluation.

compositions without additional thermodynamic calculations. Data quality is enhanced by applying the Filtering by Assessed Ternaries (FAT) criterion to ensure thermodynamic credibility, with multiple FAT thresholds examined to assess sensitivity of the results to this parameter. Demonstrated for FCC phase prediction in a ten-element HEA system, this approach should be readily adaptable to other alloy systems and phases, providing an efficient framework for exploring composition-phase relationships.

2. Data generation

The thermodynamic dataset was generated using Thermo-Calc® 2023b software coupled with the TCHEA5 thermodynamic database. Computational automation and data extraction were implemented via TCPython module, a Python-based SDK that provides a direct interface to Thermo-Calc and enables integration with numerical libraries. Simulations were performed at 900 °C under global equilibrium conditions, considering only stable phases. The compositional space comprised ten elements (Al, Co, Cr, Cu, Fe, Ni, Mn, Mo, Ti, and V), each varied from 0 to 100 at.% in 10 at.% increments, with nickel serving as the balance element to maintain a total of 100 at.%. This sampling yielded 92,377 unique alloy compositions. For each composition/temperature pair, the calculations returned the identities of the stable phases, their molar fractions, and the equilibrium compositions of each phase in atomic percent.

Of these, 72530 (78.5%) converged successfully, while 19,848 (21.5%) did not reach convergence within the allowed number of iterations of the Gibbs energy minimization algorithm. To assess whether these failures introduce systematic bias into the dataset, a comprehensive characterization of the failed compositions was performed. A nearest-neighbor analysis within the compositional grid showed that 80.8% of failed points have no failed neighbors within one grid step in any compositional direction, with a mean of 0.4 failed neighbors per point, indicating that failures are scattered across the space rather than concentrated in contiguous compositionally distinct regions. Element-wise, convergence failures were distributed uniformly across nine of the ten elements, with failure rates ranging from 22.1% to 23.8%,

consistent with the overall failure rate of 21.5%, indicating no elemental preference for failure. The exception is Ni, which showed a markedly lower failure rate of 3.3%. Complementary analysis of the mean elemental composition of failed versus converged points revealed that failed compositions present a mean Ni content of 1.77 at.%, compared to 12.25 at.% in converged points, indicating that failures are concentrated in compositions where other elements collectively dominate and Ni acts as a near-zero balance. Furthermore, applying a FAT (Fraction of Assessed Ternaries) threshold of 40% independently excludes 62.1% of the failed compositions, further reducing their impact on the final dataset. Taken together, these analyses indicate that convergence failures do not introduce a meaningful systematic bias and do not compromise the suitability of the dataset for the proposed machine learning framework.

3. Data filtering

To quantify the thermodynamic credibility of each composition, the Fraction of Assessed Binaries (FAB) and Fraction of Assessed Ternaries (FAT) metrics of Senkov et al. [21,22] were applied. FAB and FAT are computed by enumerating all binary and ternary subsystems associated with a given composition and determining what fraction of them is critically assessed in the thermodynamic database. Because TCHEA5 includes full coverage of all binaries in the ten-element space, filtering depended exclusively on FAT. Of the 120 possible ternary subsystems within this element set, only 45 are critically assessed in TCHEA5, reflecting the current state of experimental coverage for these systems. Thresholds from 0 to 100% (in 10% increments) were examined to characterize the trade-off between data quality and quantity (Fig. 2).

At the dataset level (Fig. 2.a), the fraction of removed compositions increases from 7.3% at FAT = 10% to 46.3% at FAT = 40%, exceeding 90% above FAT = 70%. A threshold of FAT = 40% was selected as it corresponds to the minimum recommended in the literature for adequate thermodynamic credibility [23], while minimizing data loss. A parallel unfiltered dataset (FAT = 0%) was retained to evaluate sensitivity to this choice and any bias introduced by the filtering.

Across system order (Fig. 2.b), ternary systems exhibit a distinct behavior, losing 62.7% of their entries regardless of threshold, a direct consequence of ternary FAT being binary in nature: a three-element system either has its sole ternary assessed or it does not. Higher-order systems are progressively eliminated above FAT = 40%, consistently faster than the dataset average, with 7- and 8-element systems losing 85.2% and 95.0% at FAT = 50%, compared to 66.4% overall. These trends indicate that stricter credibility criteria disproportionately affect higher-order, chemically complex systems.

Element-wise filtering (Fig. 2.c) shows that V-, Mn-, and Mo-containing alloys are consistently the most affected, with 66.3%, 58.0% and 55.7% at FAT = 40%, reflecting the near-complete absence of critically assessed ternaries involving these elements in TCHEA5. Fe-, Ni-, and Ti-containing alloys show the lowest removal rates (34.3%, 37.7%, and 38.9%). This elemental bias is an inherent consequence of the current state of thermodynamic database development, and its impact on model performance is explicitly evaluated through the parallel FAT = 0% dataset.

4. Data augmentation

Each composition was assigned a categorical label, hereafter termed *phase_score*, which encodes the phase constitution at 900 °C as determined by the CALPHAD calculation. The labeling scheme classified alloys into three categories: single-phase disordered FCC (class 1); disordered FCC coexisting with one or more secondary phases (class 2); and all other phase assemblages, including ordered L12 and FCC-free compositions (class 3). This three-class scheme enables machine learning models to discriminate between pure and multiphase states while retaining interpretability with respect to the main structural

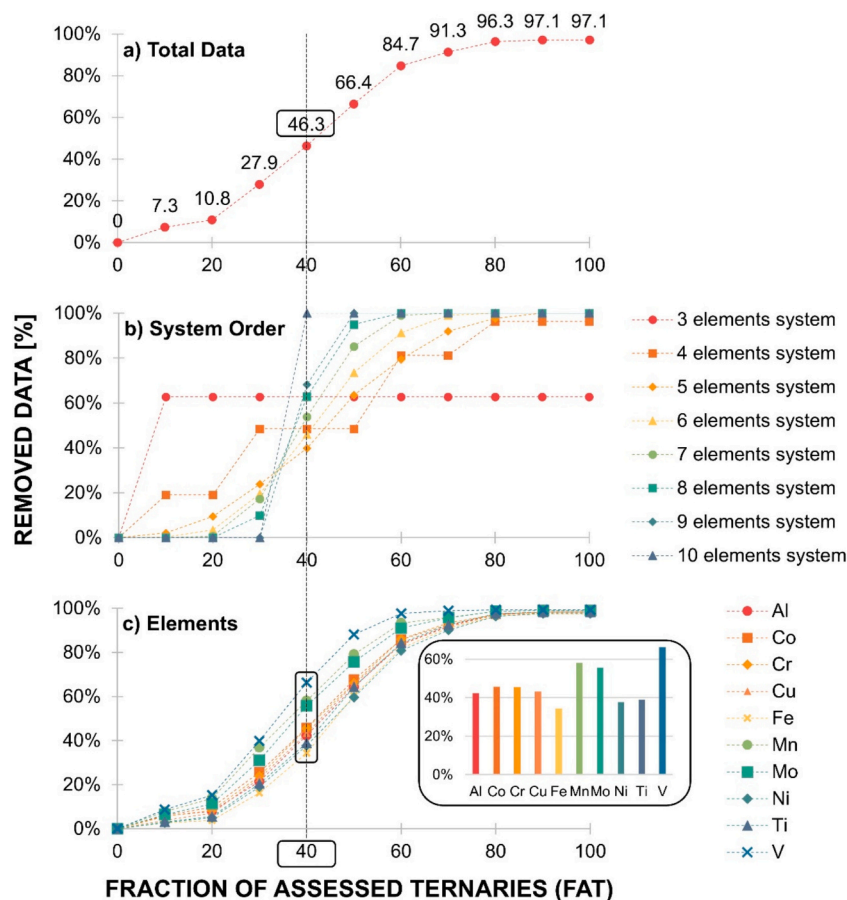


Fig. 2. - Effect of the FAT threshold on dataset composition. a) Fraction of total compositions removed as a function of FAT, showing a progressive increase in data loss with higher thresholds. b) Percentage of removed compositions grouped by system order (number of constituent elements), highlighting the disproportionate elimination of higher-order systems at stringent FAT levels. c) Element-specific fraction of excluded compositions, indicating that Mn, Mo, and V-containing alloys are most impacted, reflecting gaps in ternary assessments within the TCHEA5 database.

archetypes in high-entropy alloys design.

To address the limited number of single-phase FCC compositions in

the dataset, a data-augmentation strategy was devised to synthetically enrich the training set while preserving thermodynamic consistency.

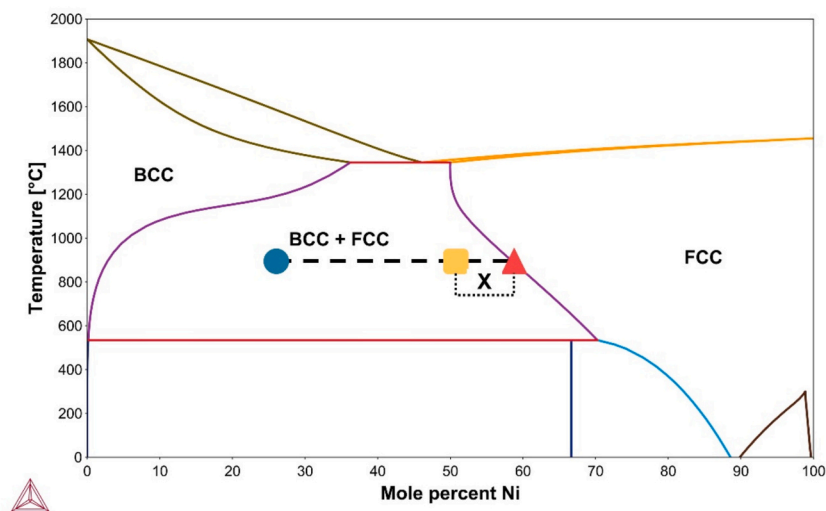


Fig. 3. - Schematic illustration of the vector-based data augmentation strategy for FCC-containing alloys, shown here on the Cr—Ni binary phase diagram for clarity (calculated with Thermo-Calc software and the TCHEA5 database). From a multiphase alloy (blue circle) containing FCC and BCC phases at 900 °C, the equilibrium composition of the FCC phase (red triangle) is extracted. A vector (dashed black line) connects the FCC endpoint to the original alloy composition. A new point (yellow square) is added at a fixed distance X along this vector, enriching the dataset with new FCC-containing entries. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Following the stratified train/test split (80/20), augmentation was applied exclusively to the training set to prevent any leakage between augmented and evaluation data. For $\text{FAT} = 0\%$ and $\text{FAT} = 40\%$, the training sets contained 58,024 and 31,183 compositions respectively, with single-phase FCC (class 1) representing only 2.3% and 2.5% of each, a severe class imbalance that constrains the performance of classification models on the minority class.

The augmentation procedure targets multiphase compositions containing at least one disordered FCC phase (class 2). Since TCHEA5 models both disordered FCC_A1 and ordered L12 under the same FCC_L12 phase model, ordered and disordered instances were distinguished by comparing sublattice site-fraction occupancies, in which equal occupancies across sublattices indicate a disordered structure, while unequal occupancies indicate L12 ordering. Only disordered FCC instances were retained as augmentation endpoints. For compositions exhibiting spinodal decomposition into two coexisting disordered FCC phases, each was treated independently, generating a separate derived endpoint per phase.

For each eligible multiphase composition (e.g., $\text{FCC} + \text{BCC}$), the equilibrium atomic composition of the disordered FCC phase, extracted directly from the CALPHAD output, defines a single-phase FCC endpoint (red triangle, Fig. 3). A vector is then constructed from this endpoint toward the original global multiphase composition (blue circle, Fig. 3), in which a synthetic point (yellow square, Fig. 3) is placed at a fixed fractional distance X . Augmentation distances (X) of 10, 30, 50, 70, and 90% were applied, producing five augmented training sets per FAT threshold in addition to the unaugmented baseline (distance = 0%).

The augmentation markedly improved class balance. For $\text{FAT} = 0\%$, the training set expanded from 58,024 to 123,292 compositions, with the single-phase FCC fraction rising from 2.3% to 27.6% (33,984 compositions). For $\text{FAT} = 40\%$, the training set expanded from 31,183 to 69,085 compositions, with the single-phase FCC fraction rising from 2.5% to 28.6% (19,728 compositions). In both cases, class 2 (FCC + secondary phases) represents approximately 48–49% of the augmented set and class 3 (no FCC) approximately 22–24%, yielding a substantially more balanced distribution across all three classes.

Beyond correcting class imbalance, the strategy densifies the sampling of the compositional space, particularly near FCC phase boundaries, which is advantageous for margin-based classifiers such as SVM by improving boundary resolution and reducing model variance. It also fills the discrete gaps inherent to the original 10 at.% grid, yielding a more continuous and representative feature space. In addition to these statistical and structural benefits, the method is also highly efficient computationally: the interpolated compositions were generated in roughly 60 s via geometric vector operations, whereas computing an equivalent number of CALPHAD equilibria on the same personal computer (Intel Core i5-10400F, 16 GB RAM) at the observed rate of ~22 compositions per minute would require approximately 28–49 h depending on the FAT threshold applied. This contrast underscores the scalability of the approach for constructing large, thermodynamically grounded datasets at minimal computational cost.

The proposed construction is physically grounded in the fundamental principles of the lever rule and the tie-line rule. In a binary two-phase region, any overall composition lying between the two equilibrium phase endpoints represents a stable mixture of those phases, with molar fractions determined by the relative distance to each endpoint. In multicomponent composition spaces, this geometric construction generalizes to tie-triangles (ternaries systems) or higher-dimensional tie-simplices ($n > 3$) [24]. Thermodynamically, this linear interpolation is valid because the molar Gibbs energy of the aggregate system lies exactly on the common tangent plane (or hyperplane) defined by the identical chemical potentials of the coexisting phases.

The thermodynamic validity of the augmentation procedure was assessed by independently recalculating representative generated compositions using Thermo-Calc with the TCHEA5 database. For the derived FCC endpoints (class 1), 97.9% of recalculated compositions returned

FCC molar fractions $\geq 99.9\%$, while only one point presented a secondary phase fraction exceeding 10%. For the interpolated compositions (class 2), phase mismatch, defined as any difference between the recalculated and original phase assemblage, was observed in approximately 11–22% of the recalculated points depending on interpolation distance. However, complete loss of FCC, corresponding to a misclassification between classes 2 and 3, occurred in fewer than 0.1% of cases.

These deviations likely reflect limitations associated with the numerical implementation of thermodynamic calculations: while the lever rule and tie-simplex construction are exact within classical thermodynamic theory, CALPHAD calculations rely on iterative Gibbs energy minimization procedures that may occasionally converge to alternative local solutions, particularly where the energy landscape is complex [24]. Nevertheless, the independent validation demonstrates that the augmentation strategy preserves the intended class assignment with high reliability across all distances examined.

5. Feature engineering

Feature engineering was used to transform raw alloy compositions into quantitative descriptors tailored for predictive modeling. In materials informatics, descriptors encode physicochemical attributes that, when appropriately selected, enhance the predictive power and generalization of machine-learning models beyond the limits of the training set. For phase-stability problems, commonly used descriptors include atomic size and bonding metrics, thermodynamic quantities, and electronic measures, enabling algorithms to capture composition-structure-property relationships [25,26]. Each composition was mapped to a vector of 20 descriptors derived from elemental properties and thermodynamic properties of mixing. Eight elemental properties: atomic radius, atomic weight, electronegativity, melting point, boiling point, valence electron concentration (VEC), enthalpy of fusion, and entropy of fusion from the Magpie library by Ward et al. [27], were aggregated per composition using atomic-fraction-weighted means and standard deviations (16 descriptors). Four additional mixture descriptors were included: atomic size mismatch (δ), configurational entropy of mixing (ideal), enthalpy of mixing (in the liquid phase), and excess heat capacity; the latter two were computed following Deffrennes et al. [28].

6. Machine learning models training

Phase classification was performed using scikit-learn, version 1.7.2 [29], with four supervised algorithms: Histogram-based Gradient Boosting (GB), Random Forest (RF), Support Vector Machine (SVM), and k-Nearest Neighbors (KNN). SVM and KNN were embedded in a pipeline with standard score normalization applied prior to training, as both algorithms are sensitive to feature scale. A majority-class dummy classifier was included as a baseline for all experiments. For each combination of FAT threshold (0 and 40%) and augmentation distance (0, 10, 30, 50, 70, and 90%), an independent model was trained and evaluated, yielding a total of 12 dataset configurations per algorithm.

Hyperparameter tuning was performed via Optuna [30], version 4.8.0, with 100 trials per dataset-model combination, using stratified 5-fold cross-validation and balanced accuracy as the optimization objective, given the unequal representation of phase classes. The search spaces covered kernel type, regularization strength (C), kernel coefficient (gamma), polynomial degree, and class weighting for SVM; number of estimators, tree depth, minimum samples per split and per leaf, feature sampling strategy, and class weighting for RF; number of iterations, learning rate, tree depth, minimum samples per leaf, and L2 regularization for GB; and number of neighbors, distance metric, distance weighting scheme, Minkowski exponent, and leaf size for KNN. After optimization, the best hyperparameters were used to retrain each model on the full training set, followed by evaluation on the held-out test set. Performance was assessed using balanced accuracy as the primary metric (Table 1), complemented by F1-macro, one-vs-rest ROC-

Table 1

- Model performance (balanced accuracy) across different FAT threshold (0 and 40%) and augmentation distance (0, 10, 30, 50, 70, and 90%), with distance = 0 corresponding to the original dataset (no extra points).

FAT = 0%						
Distance	0	10	30	50	70	90
Model						
KNN	91.8	92.9	93.6	94.0	93.8	93.5
RF	94.3	94.5	95.0	95.2	95.3	95.4
GB	93.7	95.7	96.2	95.9	95.5	95.2
SVM	95.8	96.1	96.2	96.2	96.2	96.8

FAT = 40%						
Distance	0	10	30	50	70	90
Model						
KNN	92.4	93.8	94.0	93.8	93.6	93.0
RF	93.9	94.9	95.4	95.3	95.4	95.1
GB	94.6	96.2	96.4	96.0	95.5	95.5
SVM	96.2	96.7	96.6	96.4	96.4	96.4

AUC and 95% bootstrap confidence intervals. Precision-recall curves were computed per class to evaluate minority-class separability. The complete sets of optimized hyperparameters for all models and datasets, along with their respective performance metrics, are available in the Supplementary Material (Tables 1–4).

Analysis of Table 1 reveals three principal trends. First, data augmentation consistently improves balanced accuracy relative to the

unaugmented baseline (distance = 0%) for all evaluated models and FAT thresholds, indicating that the generated compositions provide useful training information. Second, the largest improvements are generally observed for intermediate interpolation distances (30–50%), while performance tends to plateau at higher distances. This behavior is consistent with the geometric rationale of the augmentation strategy: points generated too close to the FCC endpoint contribute limited additional information, whereas points generated at larger interpolation distances progressively approach the original multiphase compositions. Third, FAT = 40% yields marginally higher balanced accuracy than FAT = 0% at equivalent augmentation distances. This may reflect the increased thermodynamic reliability associated with the FAT filtering procedure, compositional redistribution caused by preferential exclusion of V-, Mn-, and Mo-containing alloys, or a combination of both.

Among the evaluated models, SVM achieved the highest balanced accuracy across most configurations, reaching 96.8% at FAT = 0% and distance = 90%, followed by GB, RF, and KNN. This consistent superiority indicates that SVM is particularly effective at capturing the composition-phase relationships represented by the selected descriptors. While SVM attained the highest absolute scores, it showed the smallest relative improvement from data augmentation (+0.2–1.0%), likely due to its already strong baseline performance on the unaugmented dataset. In contrast, GB and KNN benefited most proportionally (+1.6–2.5%), though overall gains remained modest (~1–3%) given the high baseline performance across all models. The impact of augmentation on minority-class performance is further examined through class-wise precision-recall analysis and confusion matrices, as presented below.

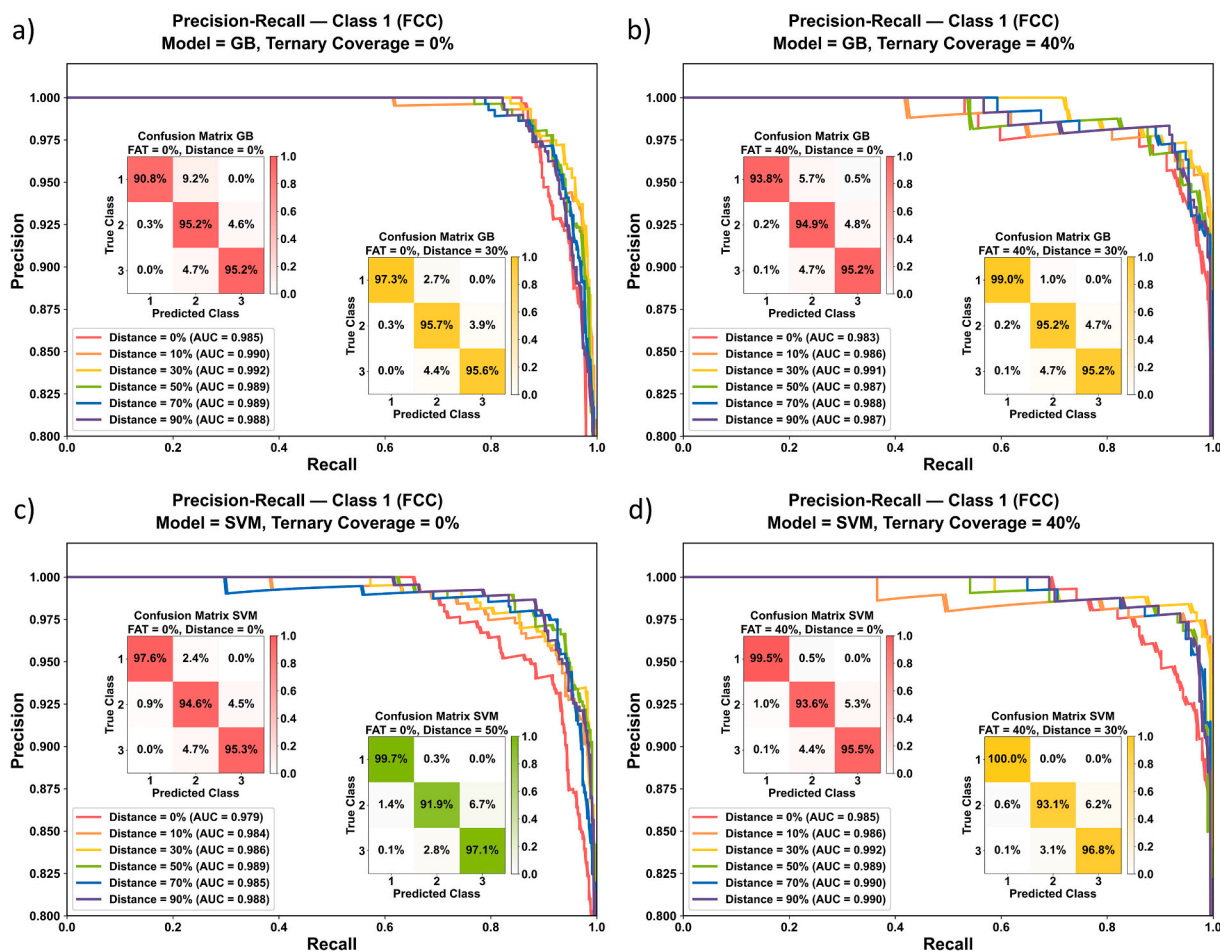


Fig. 4. - Precision-recall curves for class 1 (single-phase FCC) and confusion matrices at distance = 0% (baseline) and distance = 30% (best-performing augmentation), for GB (a, b) and SVM (c, d) under FAT = 0% and FAT = 40%. Precision-recall curves show all augmentation distances; confusion matrices highlight the baseline and the configuration yielding the highest AUC.

Fig. 4 presents the precision-recall curves for class 1 (single-phase FCC) along with the corresponding confusion matrices for the baseline condition (distance = 0%) and representative augmentation conditions associated with the highest AUC-PR values for the GB and SVM models. These models were selected due to their consistently high balanced accuracy; complete results for all configurations are provided in the Supplementary Material (Figs. S1-S2). Since class 1 represents a minority fraction of the dataset, precision-recall analysis provides a more informative assessment than ROC-based metrics. For GB, the highest AUC-PR values are obtained at distance = 30%, increasing from 0.985 to 0.992 at FAT = 0% and from 0.983 to 0.991 at FAT = 40%. For SVM, the highest values occur at distance = 50% (FAT = 0%) and distance = 30% (FAT = 40%), increasing from 0.979 to 0.989 and from 0.985 to 0.992, respectively.

The confusion matrices provide a direct visualization of these improvements in minority-class recognition. For GB, class 1 recall increases from 90.8% to 97.3% under FAT = 0% and from 93.8% to 99.0% under FAT = 40%, corresponding to gains of +6.5 and +5.2 percentage points, respectively. For SVM, recall improves from 97.6% to 99.7% under FAT = 0% and from 99.5% to 100.0% under FAT = 40%. Crucially, these improvements are accompanied by only minor changes in the classification of classes 2 and 3, indicating that the augmentation strategy enhances minority-class recognition without compromising majority-class performance.

Methodology Applicability.

Although demonstrated here for FCC phase prediction, the proposed augmentation strategy is not inherently tied to a specific crystal structure. The lever-rule construction requires only that the target phase appears in multiphase CALPHAD equilibria with an extractable equilibrium composition. To illustrate this transferability, the same pipeline was applied to BCC phase prediction using the same ten-element database, the same FAT thresholds (0 and 40%), the most promising augmentation distance (30%), and the two best-performing models (GB and SVM).

In this new dataset, single-phase BCC (class 1) represented 4.1% and 3.7% of the training set under FAT = 0% and FAT = 40% respectively, a less severe but still meaningful class imbalance. The augmentation expanded the training sets from 58,024 to 200,782 compositions (FAT = 0%) and from 31,183 to 98,471 (FAT = 40%), with class 1 fractions rising to 36.7% and 35.3% respectively. Machine learning results are presented below (Table 2, Fig. 5).

Table 2 and Fig. 5 demonstrate that the proposed augmentation strategy is transferable to BCC phase prediction despite the distinct phase relationships relative to the FCC case. In all evaluated configurations, data augmentation improved balanced accuracy relative to the original dataset. For GB, balanced accuracy increased from 88.05% to 90.81% under FAT = 0% and from 84.49% to 88.04% under FAT = 40%. SVM again achieved the highest overall performance, improving from 93.90% to 94.40% under FAT = 0% and from 92.50% to 93.43% under FAT = 40%.

Fig. 5 further evaluates minority-class recognition through precision-recall analysis for class 1 (single-phase BCC). In all evaluated cases,

Table 2

BCC Model performance across different FAT threshold (0 and 40%) and augmentation distance (0 and 30%), with distance = 0 corresponding to the original dataset (no extra points).

Model	FAT (%)	Distance (%)	Balanced Accuracy (%)
GB	0	0	88.05
		30	90.81
	40	0	84.49
		30	88.04
SVM	0	0	93.90
		30	94.40
	40	0	92.50
		30	93.43

AUC-PR increased following augmentation, indicating improved discrimination of the minority BCC class. The confusion matrices additionally show substantial improvements in class 1 recall for GB, increasing from 82.4% to 89.8% under FAT = 0% and from 71.1% to 80.8% under FAT = 40%. For SVM under FAT = 40%, class 1 recall also improved slightly, from 96.9% to 97.2%. A small decrease was observed for SVM under FAT = 0% (99.2% to 98.8%) despite the improved AUC-PR after augmentation.

Overall, these results indicate that the proposed augmentation strategy is not restricted to FCC prediction and can be successfully transferred to other phase-stability classification problems within the same CALPHAD-informed framework.

7. Limitations

Apart from its benefits, the methodology presents inherent limitations worth mentioning. First, the approach relies entirely on CALPHAD equilibrium calculations, whose reliability depends on the quality and completeness of the underlying binary and ternary assessments, the ability of the software to find the true global energy minimum, and the assumption of full thermodynamic equilibrium. Such equilibrium may not be achieved in experimentally processed alloys due to kinetic constraints, metastable phases, or short-range ordering effects not captured by classical CALPHAD models. Second, transferability to phases such as HCP or intermetallics may require a dedicated compositional database constructed around elements that stabilize those structures. In the present ten-element database, single-phase HCP compositions represent only 0.03% of all converged equilibria (20 out of 72,530), with only 536 compositions eligible for augmentation, insufficient for robust model training. Third, the method is temperature-specific: the lever rule and tie-simplex construction are valid only at the temperature at which the original CALPHAD equilibria were calculated, and application to other temperatures requires a new grid calculation.

Despite these limitations, the methodology demonstrates broad applicability within its defined scope: any alloy system for which a CALPHAD database exists, at any temperature for which equilibrium calculations are performed, and for any phase that appears in multiphase equilibria with extractable composition data. Within these conditions, the lever-rule augmentation strategy provides a computationally efficient, thermodynamically grounded solution to the class imbalance problem, contributing to more accurate and reliable machine learning models for alloy design.

8. Conclusion

In conclusion, this work presents a thermodynamically grounded machine learning framework for accelerated phase prediction in complex alloy systems. By combining high-throughput CALPHAD-generated data (> 72,000 compositions) with lever-rule-informed data augmentation, the proposed methodology efficiently addresses class imbalance. Crucially, it expands the dataset by over 100% via geometric vector operations in just 60 s, bypassing 28–49 h of additional equilibrium calculations.

The augmentation strategy consistently improved balanced accuracy and minority-class recognition across all models, with SVM achieving the highest baseline performance. Although global performance gains were modest (1–3%), substantial improvements of up to 6.5 percentage points in recall were achieved for the minority single-phase FCC class. Furthermore, independent thermodynamic validation confirmed that 99% of the derived single-phase endpoints retained FCC fractions $\geq 99\%$, proving high physical reliability.

Beyond FCC prediction, successful application to BCC phase classification demonstrated that the methodology is highly transferable to other crystal structures. Overall, the proposed approach provides a scalable, computationally efficient, and physically interpretable strategy for integrating thermodynamic modeling and machine learning in alloy

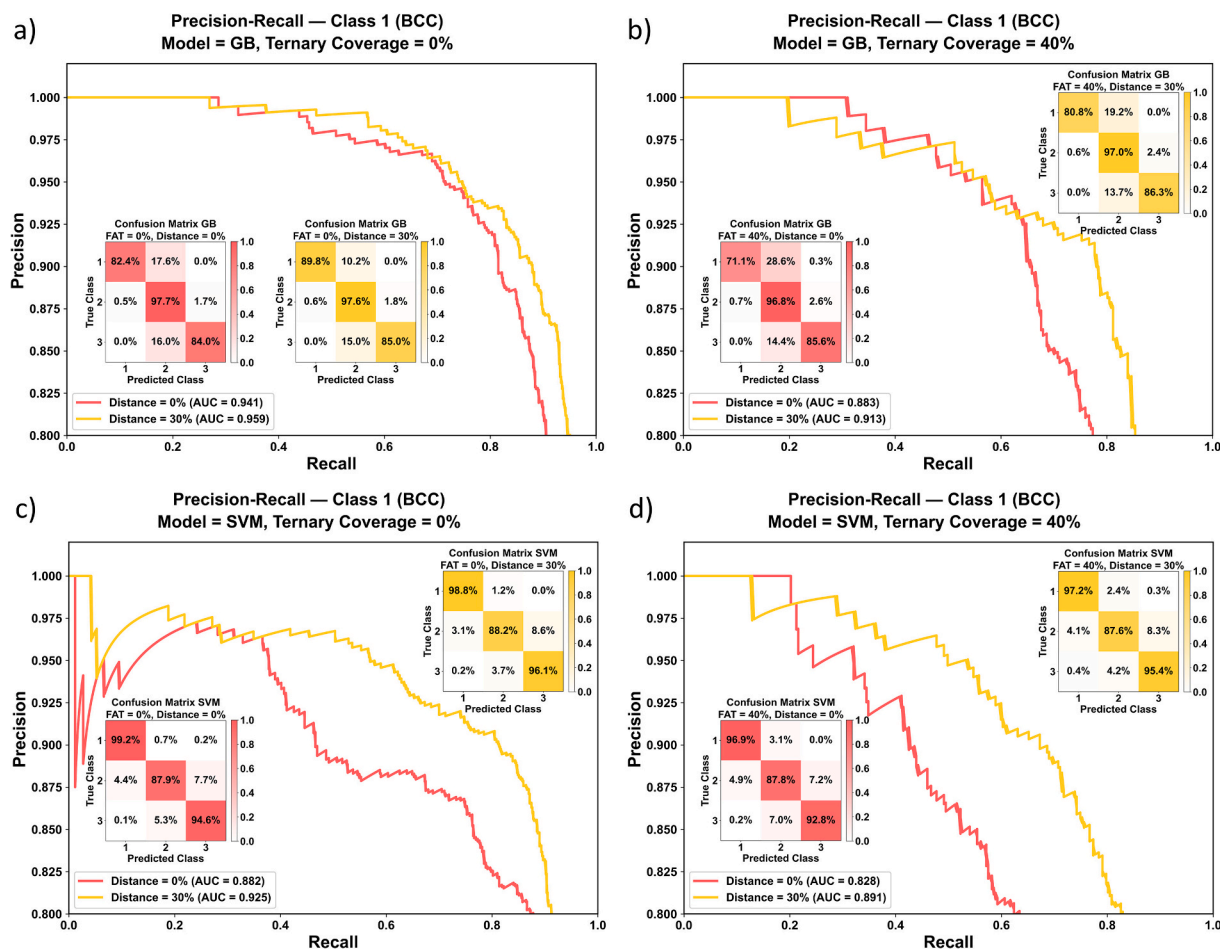


Fig. 5. Precision-recall curves for class 1 (single-phase BCC) together with the corresponding confusion matrices for the GB and SVM models under FAT = 0% and FAT = 40%. Curves are shown for the original dataset (distance = 0%) and for the augmented dataset at distance = 30%.

design.

CRediT authorship contribution statement

Caroline Binde Stoco: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guillaume Deffrennes:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Yannick Champion:** Writing – review & editing, Supervision, Conceptualization. **Francisco Gil Coury:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001; Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) [grant numbers 2023/04907-2, 2023/14613-6].

Data availability

Code and processed data are available at <https://github.com/Caroline-B-S/Data-Augmentation-CALPHAD-ML>. Raw CALPHAD data cannot be shared due to commercial licensing restrictions.

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